

## Author

Ruchika Narendra Patil

23f2002484

23f2002484@ds.study.iitm.ac.in

I'm a friendly and curious individual who loves exploring new things. I enjoy solving problems, collaborating with others and continuously improving my skills.

## Description

This project is about creating a quiz app where an admin manages subjects, chapters, quiz and questions, while users can sign up, take quizzes and see their scores.

## Technologies used

Backend:

- Flask: Lightweight web framework for routing and session management.
- Flask-SQLAlchemy: ORM for seamless database interactions.
- SQLAlchemy: Manages database queries and relationships.
- pytz: Handles time zone consistency (IST).
- datetime: For date and time operations.
- SQL functions (func): Enables SQL aggregation and calculations.

Frontend:

- HTML5 + CSS3: For structuring and styling web pages.
- Bootstrap: Provides responsive UI components.
- Jinja2: Template engine for dynamic content rendering.

Database:

- SQLite: Lightweight, file-based relational database.

## DB Schema Design

### 1. User Class

- id: User ID, integer, primary key, auto-increment.
- email: Email, string (100), unique, cannot have null value.
- password: Password, string (20), cannot have null value.
- username: Username, string (100), cannot have null value.
- qualification: Qualification, string (100), cannot have null value.
- date\_of\_birth: Date of birth, string (20), cannot have null value.
- role: User role, string (20), cannot have null value, default = "Student".

### 2. Subject Class

- id: Subject ID, string (50), primary key.
- name: Subject name, string (100), cannot have null value.
- description: Description, text, cannot have null value.

### 3. Chapter Class

- id: Chapter ID, string (50), primary key.
- name: Chapter name, string (100), cannot have null value.
- description: Chapter description, text, cannot have null value.
- subject\_id: Foreign key, string (100), references id in the Subject table.
- Relationship: One-to-many relationship with Subject (a subject can have multiple chapters).

### 4. Quiz Class

- id: Quiz ID, string (50), primary key.
- chapter\_id: Foreign key, string (100), references id in the Chapter table.
- remarks: Quiz remarks, string (100), cannot have null value.
- quiz\_date: Quiz date, date, cannot have null value.
- time\_dur: Quiz duration, string (5), cannot have null value.
- Relationship: One-to-many relationship with Chapter (a chapter can have multiple quizzes).

### 5. Question Class

- id: Question ID, string (50), primary key.
- quiz\_id: Foreign key, string (100), references id in the Quiz table.
- que: Question text, text, cannot have null value.
- o\_a: Option A, string (300), cannot have null value.
- o\_b: Option B, string (300), cannot have null value.
- o\_c: Option C, string (300), cannot have null value.
- o\_d: Option D, string (300), cannot have null value.
- correct\_o: Correct option, string (1), cannot have null value.
- Relationship: One-to-many relationship with Quiz (a quiz can have multiple questions).

### 6. Score Class

- id: Score ID, integer, primary key, auto-increment.
- quiz\_id: Foreign key, string (100), references id in the Quiz table.
- user\_id: Foreign key, string (100), references id in the User table.
- attempt\_timing: Attempt time, datetime, default = current time in IST.
- tot\_score: Total score, integer, cannot have null value.
- Relationship:
  - One-to-many relationship with Quiz (a quiz can have multiple scores).
  - One-to-many relationship with User (a user can have multiple scores).

ER Diagram :

<https://drive.google.com/file/d/14t8drDuQCHGkEzfXSrfjobif8ekdUeTV/view?usp=sharing>

## Architecture and Features

Architecture :

1. `app.py`:
  - Main application file.
  - Handles routes, request processing, and rendering templates.
2. `models.py`:
  - Defines the database schema.
3. `instance` folder:
  - Contains the database.
4. `static` folder :
  - Holds static assets( images).
5. `templates` folder:
  - Contains HTML files for rendering the frontend.
6. `venv`:
  - Virtual environment for managing project dependencies locally.

Features :

Admin Functionality:

Login: Admin logs in using a pre-existing account stored in the database.

The Admin section consists of four main options:

1. Home
  - Displays all subjects in the system.
  - Admin can create, edit, and delete subjects.
  - Clicking on a manage chapter allows the admin to manage chapters related to that subject.
  - From the chapter management page, the admin can add, modify, or remove chapters under a subject.
2. Quiz
  - Displays a list of all quizzes created in the system.
  - Admin can create, edit, or delete quizzes.
  - Clicking on a manage question allows the admin to manage questions related to that quiz.
  - From the question management page, the admin can add, modify, or remove MCQ questions (one correct option per question).
  - Admin can also set quiz dates and remarks.
3. Manage Users
  - Displays a list of all registered users in the system.
4. Summary
  - Displays charts and reports.

Search Bar : Search for users, subjects, and quizzes.

## User Functionality:

### Login & Registration:

- Users can register with their details (email, password, full name, qualification, and DOB).
- They can log in with their credentials to access the platform.

The User section consists of three main options:

#### 1. User Dashboard

- Displays the count of quiz attempts made by the user.
- Shows Success rate
- Includes a quiz history section that lists:
  - Quiz Id
  - Scores
  - Dates of previous attempts

#### 2. Quiz

- Displays upcoming quizzes that the user can attempt.
- Includes filter options to refine quizzes by Subjects and Chapters
- Users can start the quiz.
- Upon submission:
  - The score is displayed.
  - Results are stored in quiz history.

#### 3. Summary

- Displays charts and reports.

## Video

[https://drive.google.com/file/d/1wbaGeyl6Su2gKSE8u5M3Jmtdyv6gT8EZ/view?usp=drive\\_link](https://drive.google.com/file/d/1wbaGeyl6Su2gKSE8u5M3Jmtdyv6gT8EZ/view?usp=drive_link)